## Practical 5 : Develop a Laravel/Django based web application.

**Initial Steps:**

**1. Check Python Version**
Python –version

**2. Check Django version**
Python -m djanog --version

**3. Create the Project Folder**
mkdir PRODUCTCRUD

**4. Enter the Project Folder**
cd PRODUCTCRUD

**5. Create the Django Project**
django-admin startproject productproject .

**6. Create the Products App**
python3 manage-py startapp products

**7. Open Visual Studio Code**
Code .

**8. Create and Apply Migrations**
python3 manage-py makemigrations
python3 manage-py migrate

**9. Create the Template Folders**
mkdir products/templates
mkdir products/templates/products

**10. Run the Application**
python3 manage.py runserver
Open the application in a browser:
http://127.0.0.1:8000/

**Code:**

**1. Register the Project app - Open productproject/settings.py and add products to INSTALLED_APPS:**

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'products',
]
```

**2. Create the Product Model - Open products/models.py:**

```
from django.db import models

# Create your models here.
class Product(models.Model):
    name = models.CharField(max_length=200)
    category = models.CharField(max_length=100)
    price = models.DecimalField(max_digits=10, decimal_places=2)
    quantity = models.IntegerField()

    def __str__(self):
        return self.name
```

**3. Create Product Form - Create products/forms.py:**

```
from django import forms
from .models import Product

class ProductForm(forms.ModelForm):
    class Meta:
        model = Product
        fields = ['name', 'category', 'price', 'quantity']
```

**4. Create CRUD View - Open products/views.py:**

```
from django.shortcuts import render, redirect, get_object_or_404
from .models import Product
from .forms import ProductForm
```

```
# Create and Read
def product_list(request):
    products = Product.objects.all()

    if request.method == 'POST':

        form = ProductForm(request.POST)

        if form.is_valid():
            form.save()
            return redirect('product_list')
    else:
        form = ProductForm()

    return render(request, 'products/product_list.html', {
        'form': form,
        'products': products
    })

# Update
def product_update(request, id):
    product = get_object_or_404(Product, id=id)

    if request.method == 'POST':
        form = ProductForm(request.POST, instance=product)

        if form.is_valid():
            form.save()
            return redirect('product_list')
    else:
        form = ProductForm(instance=product)

    return render(request, 'products/product_update.html', {
    'form': form
    })

# Delete
def product_delete(request, id):
    product = get_object_or_404(Product, id=id)

    if request.method == 'POST':
        product.delete()
        return redirect('product_list')

    return render(request, 'products/product_delete.html', {
        'product': product
```

```
    })
```

**5. Create Application URLs - Create products/urls.py:**

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.product_list, name='product_list'),
    path('update/<int:id>/', views.product_update, name='product_update'),
    path('delete/<int:id>/', views.product_delete, name='product_delete'),
]
```

**6. Create the Product List Page - Create products/templates/products/product_list.html:**

```
<!doctype html>
<html>
 <head>
  <title>Product Management</title>
 </head>
 <body>
  <h1>Product Management</h1>
  <h2>Insert Product</h2>
  <form method="POST">
   {% csrf_token %} {{ form.as_p }}
   <button type="submit">Insert Product</button>
  </form>
  <hr />
  <h2>Product List</h2>
  {% for product in products %}
  <p>
   <b>Name:</b> {{ product.name }}<br />
   <b>Category:</b> {{ product.category }}<br />
   <b>Price:</b> ₹{{ product.price }}<br />
   <b>Quantity:</b> {{ product.quantity }}
  </p>
  <a href="{% url 'product_update' product.id %}">Update</a>
  |
  <a href="{% url 'product_delete' product.id %}">Delete</a>
  <hr />
  {% empty %}
  <p>No products available.</p>
  {% endfor %}
 </body>
</html>
```

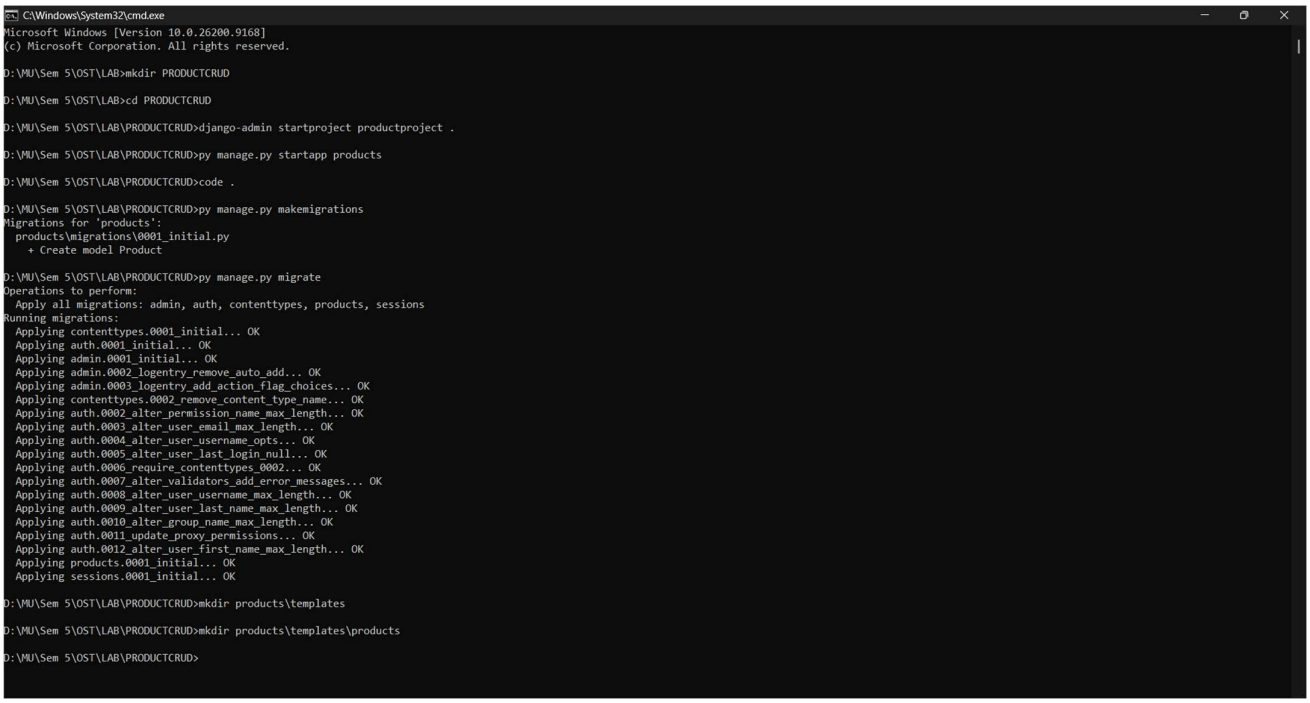
**7. Create the Update Page - Create products/templates/products/product_update.html:**

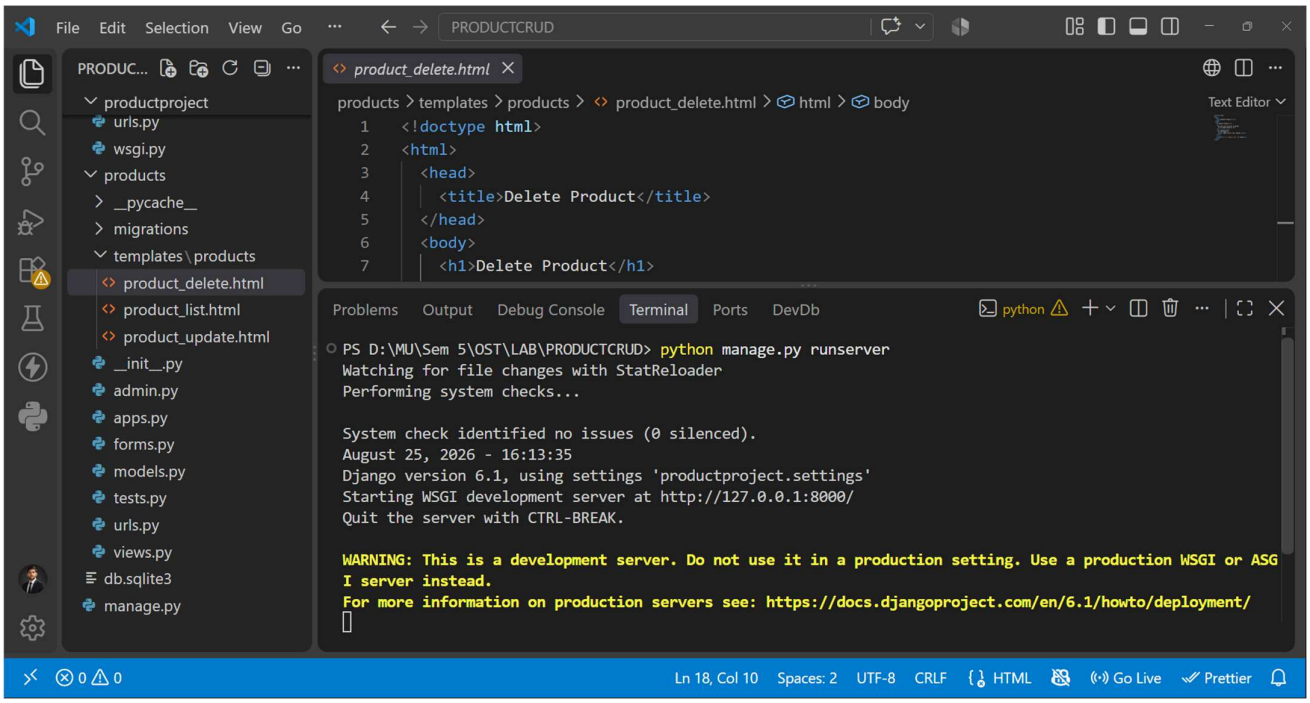
```
<!doctype html>
<html>
 <head>
  <title>Update Product</title>
 </head>
 <body>
  <h1>Update Product</h1>
  <form method="POST">
   {% csrf_token %} {{ form.as_p }}
   <button type="submit">Update Product</button>
  </form>
  <br />
  <a href="{% url 'product_list' %}">Back to Product List</a>
 </body>
</html>
```

**8. Create the Delete Page - Create products/templates/products/product_delete.html:**

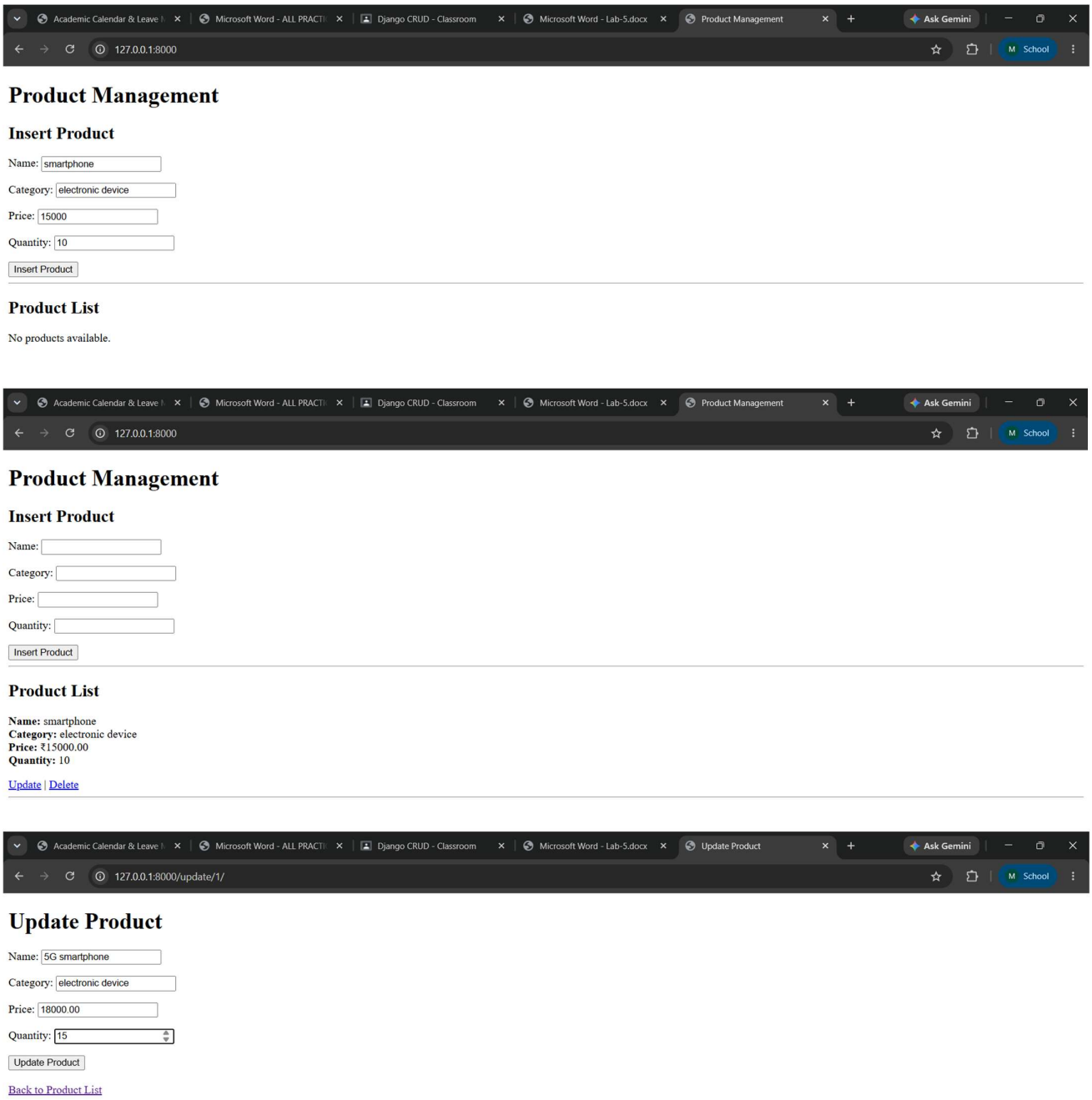
```
<!doctype html>
<html>
 <head>
  <title>Delete Product</title>
 </head>
 <body>
  <h1>Delete Product</h1>
  <p>
   Are you sure you want to delete
   <b>{{ product.name }}</b>?
  </p>
  <form method="POST">
   {% csrf_token %}
   <button type="submit">Yes, Delete</button>
  </form>
  <br />
  <a href="{% url 'product_list' %}">Cancel</a>
 </body>
</html>
```

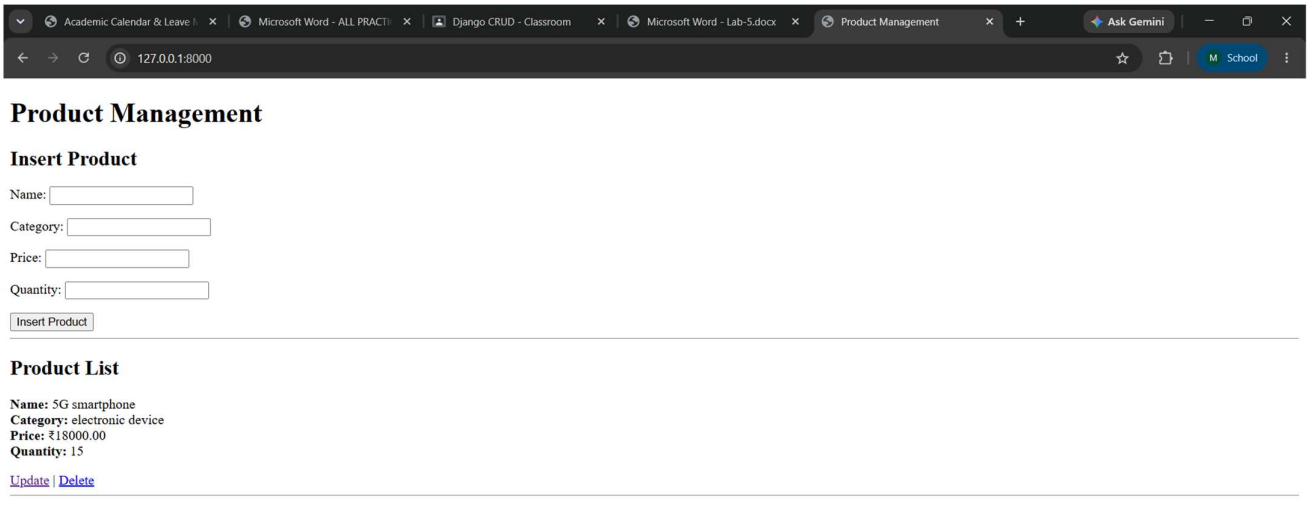
## Commands Screenshot :

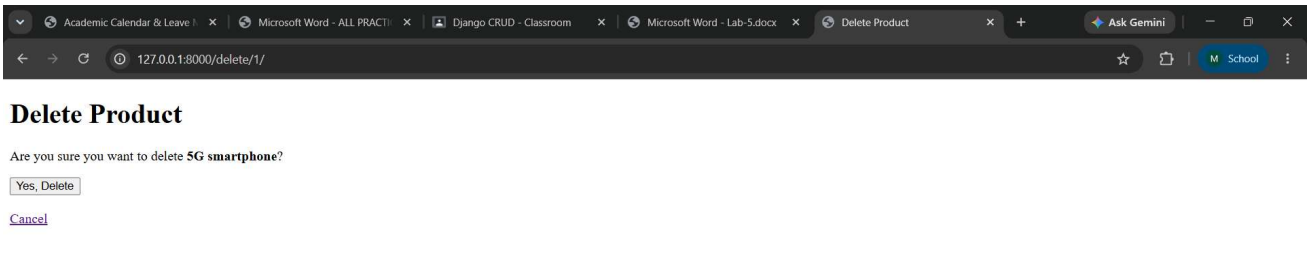


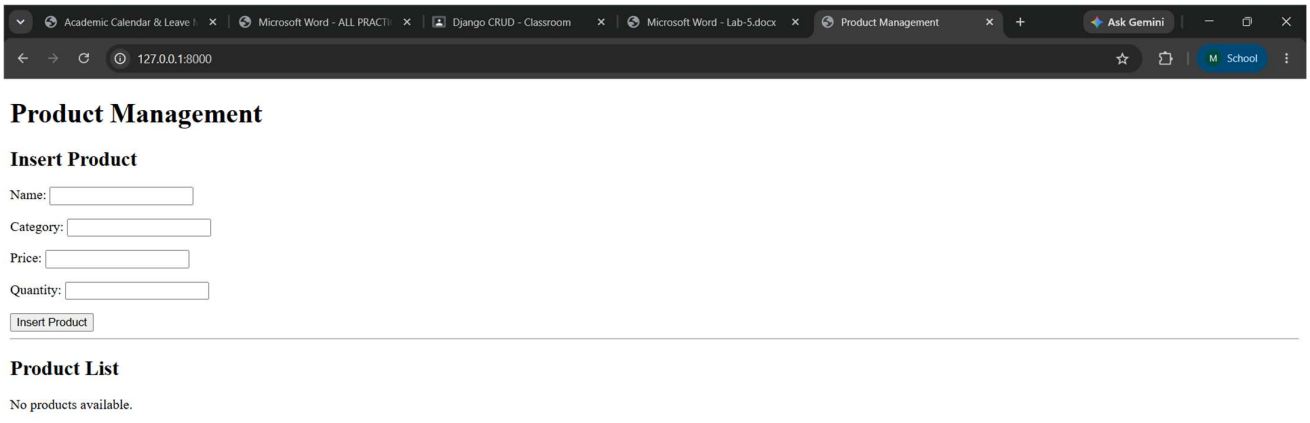



**Output:**


Academic Calendar & Leave
Microsoft Word - ALL PRACTI
Django CRUD - Classroom
Microsoft Word - Lab-5.docx
Product Management
Ask Gemini
127.0.0.1:8000
School
Product Management
Insert Product
Name: smartphone
Category: electronic device
Price: 15000
Quantity: 10
Insert Product
Product List
No products available.
Product Management
Insert Product
Name:
Category:
Price:
Quantity:
Insert Product
Product List
Name: smartphone
Category: electronic device
Price: ₹15000.00
Quantity: 10
Update | Delete
Update Product
127.0.0.1:8000/update/1/
Update Product
Name: 5G smartphone
Category: electronic device
Price: 18000.00
Quantity: 15
Update Product
Back to Product List


Academic Calendar & Leave
Microsoft Word - ALL PRACTI
Django CRUD - Classroom
Microsoft Word - Lab-5.docx
Product Management
Ask Gemini
127.0.0.1:8000
School
Product Management
Insert Product
Name:
Category:
Price:
Quantity:
Insert Product
Product List
Name: 5G smartphone
Category: electronic device
Price: ₹18000.00
Quantity: 15
Update | Delete


Academic Calendar & Leave
Microsoft Word - ALL PRACTI
Django CRUD - Classroom
Microsoft Word - Lab-5.docx
Delete Product
Ask Gemini
127.0.0.1:8000/delete/1/
School
Delete Product
Are you sure you want to delete 5G smartphone?
Yes, Delete
Cancel


Academic Calendar & Leave
Microsoft Word - ALL PRACTI
Django CRUD - Classroom
Microsoft Word - Lab-5.docx
Product Management
Ask Gemini
127.0.0.1:8000
School
Product Management
Insert Product
Name:
Category:
Price:
Quantity:
Insert Product
Product List
No products available.